ER/EER Diagram for a Corporate Project Management System

Bachelor of Technology
Computer Science and Engineering

Submitted By

RAGYAM SINHA
13000123064

JANUARY 2026

Techno Main
EM-4/1, Sector-V, Salt Lake
Kolkata- 700091
West Bengal
India

Problem Understanding & Assumptions: -

Overview:

The system manages a corporate environment with a diverse workforce working on various projects controlled by departments. The workforce is categorized into three distinct types with specific attributes.

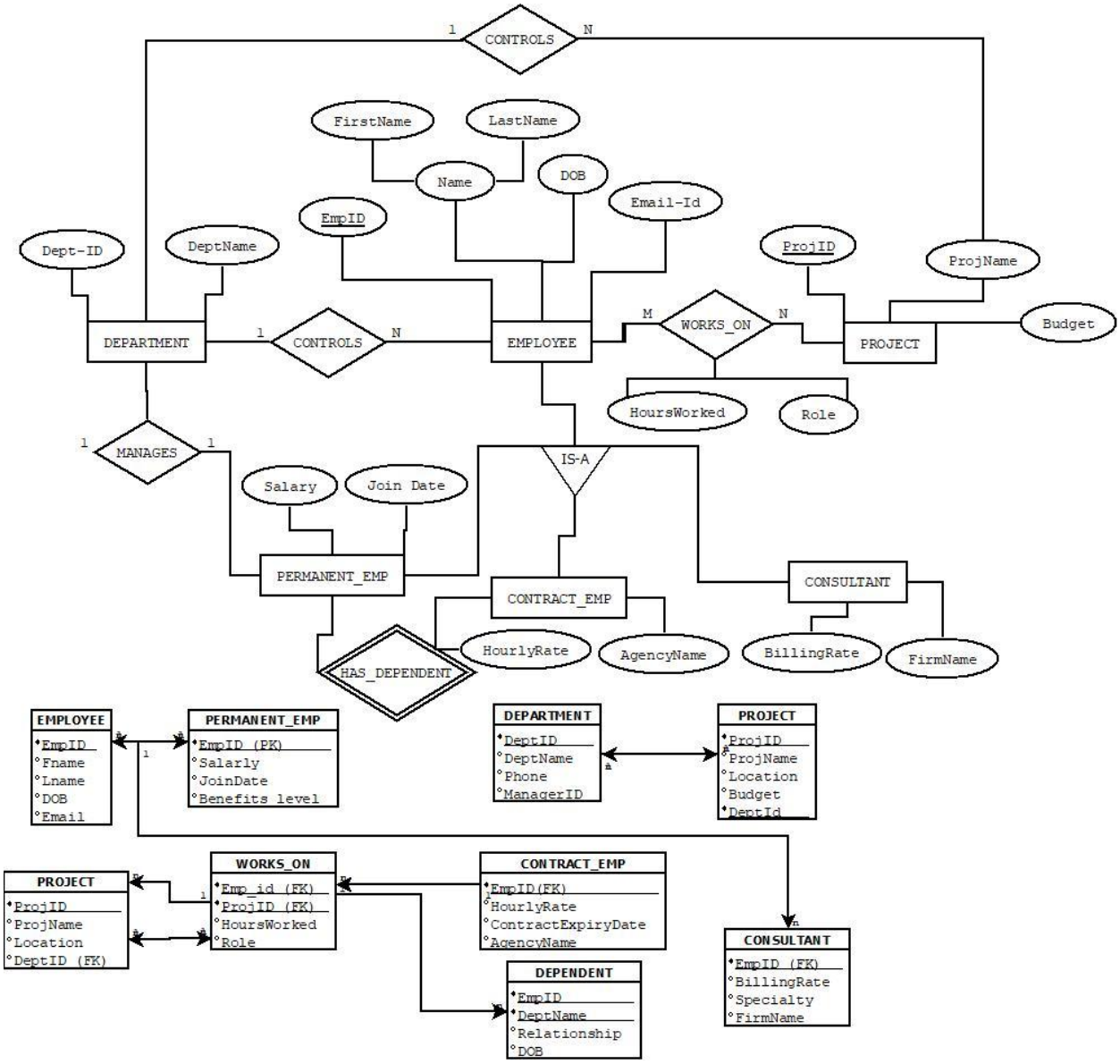
Assumptions:

1. **Hierarchy:** The categorization of employees is **Disjoint and Total**; an employee cannot belong to multiple categories simultaneously and must belong to at least one.
2. **Management:** Only **Permanent Employees** are eligible to manage Departments.
3. **Dependents:** Benefits for dependents are provided strictly to Permanent Employees.
4. **Projects:** A project belongs to exactly one department but can have multiple employees working on it.
5. **Uniqueness:** Email addresses and Department names must be unique across the system.

2. Identification of Entities & Attributes

- **EMPLOYEE:** EmpID (Key), Fname, Lname, DOB, Email.
- **PERMANENT_EMP:** Salary, JoinDate, Benefits_Level.
- **CONTRACT_EMP:** HourlyRate, ContractExpiryDate, AgencyName.
- **CONSULTANT:** BillingRate, Specialty, FirmName.
- **DEPARTMENT:** DeptID (Key), DeptName, Phone.
- **PROJECT:** ProjID (Key), ProjName, Location, Budget.
- **DEPENDENT (Weak):** Name (Partial Key), Relationship, DOB.

3. ER/EER Diagram Logic



- **Superclass/Subclass:** EMPLOYEE connects to PERM, CONT, and CONS via a circle marked "d" (disjoint) and double lines (total participation).

- # Relationships:

  - PERMANENT_EMP **MANAGES** (1,1) DEPARTMENT.
  - DEPARTMENT **CONTROLS** (1,N) PROJECT.
  - EMPLOYEE **WORKS_ON** (M,N) PROJECT (Attribute: Hours, Role).
  - PERMANENT_EMP **HAS_DEPENDENT** (1,N) DEPENDENT (Weak relationship, double diamond).

4. Relational Tables with Constraints

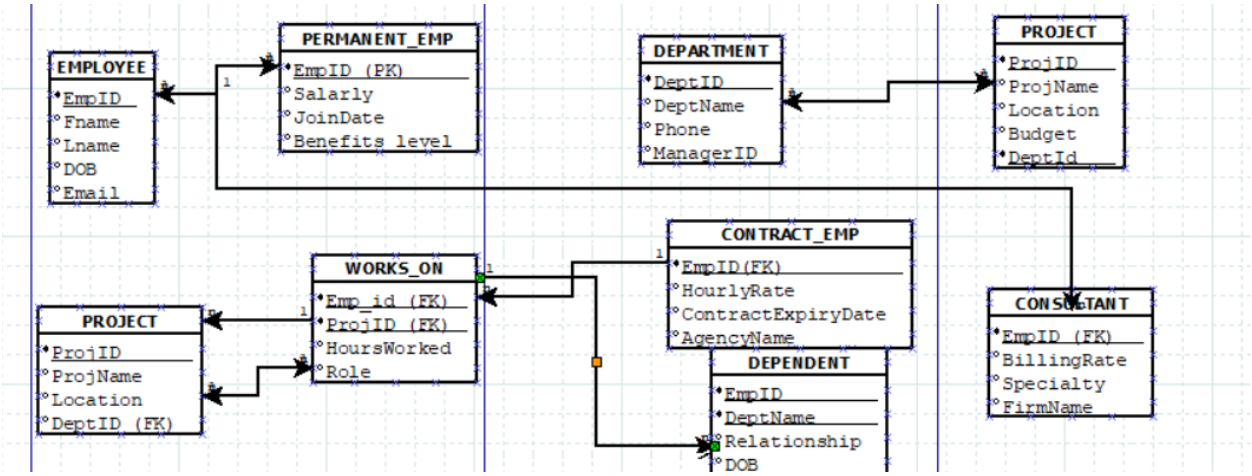


Table 1: EMPLOYEE

- **PK:** EmpID
- **FK:** None
- **Constraints:** Email UNIQUE, EmpID NOT NULL.

Table 2: DEPARTMENT

- **PK:** DeptID
- **FK:** ManagerID references PERMANENT_EMP(EmpID)
- **Constraints:** DeptName UNIQUE, ManagerID NOT NULL (Total participation).

Table 3: PROJECT

- **PK:** ProjID
- **FK:** DeptID references DEPARTMENT(DeptID)
- **Constraints:** DeptID NOT NULL (Every project must have a department).

Table 4: PERMANENT_EMP

- **PK:** EmpID
- **FK:** EmpID references EMPLOYEE(EmpID)

- **Constraints:** Salary > 0.

Table 5: CONTRACT_EMP

- **PK:** EmpID
- **FK:** EmpID references EMPLOYEE(EmpID)
- **Constraints:** HourlyRate > 0.

Table 6: CONSULTANT

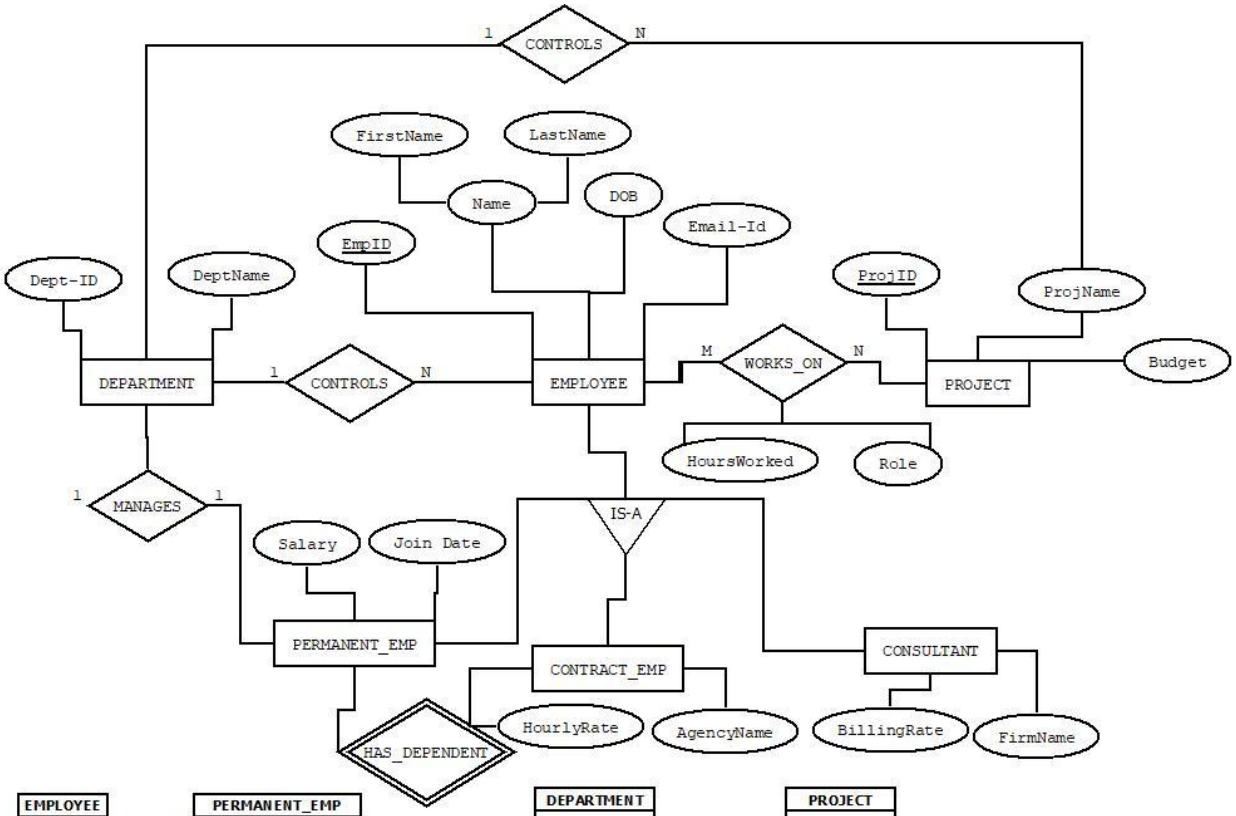
- **PK:** EmpID
- **FK:** EmpID references EMPLOYEE(EmpID)
- **Constraints:** None specific.

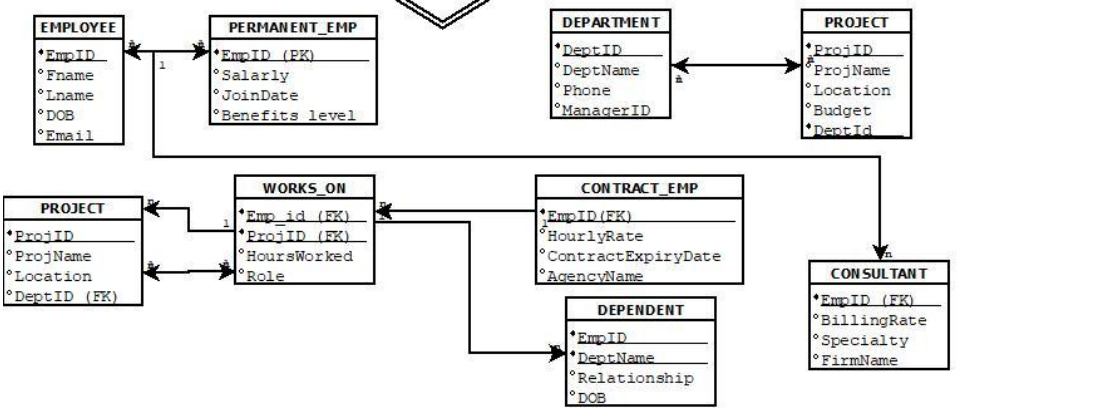
Table 7: WORKS_ON

- **PK:** (EmpID, ProjID)
- **FK1:** EmpID references EMPLOYEE(EmpID)
- **FK2:** ProjID references PROJECT(ProjID)
- **Constraints:** HoursWorked >= 0.

Table 8: DEPENDENT

- **PK:** (EmpID, DepName)
- **FK:** EmpID references PERMANENT_EMP(EmpID) (On Delete Cascade).


CONTROLS
1
N
FirstName
LastName
Name
DOB
Email-Id
EmpID
Dept-ID
DeptName
ProjID
ProjName
DEPARTMENT
CONTROLS
EMPLOYEE
WORKS_ON
PROJECT
Budget
M
N
HoursWorked
Role
MANAGES
IS-A
Salary
Join Date
PERMANENT_EMP
CONTRACT_EMP
CONSULTANT
HAS_DEPENDENT
HourlyRate
AgencyName
BillingRate
FirmName


EMPLOYEE
EmpID
Fname
Lname
DOB
Email
PERMANENT_EMP
EmpID (FK)
Salarly
JoinDate
Benefits level
DEPARTMENT
DeptID
DeptName
Phone
ManagerID
PROJECT
ProjID
ProjName
Location
Budget
DeptId
WORKS_ON
Emp_id (FK)
ProjID (FK)
HoursWorked
Role
CONTRACT_EMP
EmpID(FK)
HourlyRate
ContractExpiryDate
AgencyName
CONSULTANT
EmpID (FK)
BillingRate
Specialty
FirmName
DEPENDENT
EmpID
DeptName
Relationship
DOB

6. Final Set of Relational Schemas

- **EMPLOYEE** ($\underline{EmpID}$, Fname, Lname, DOB, Email)
- **PERMANENT_EMP** ($\underline{EmpID}$, Salary, JoinDate, Benefits_Level)
- **CONTRACT_EMP** ($\underline{EmpID}$, HourlyRate, ContractExpiryDate, AgencyName)
- **CONSULTANT** ($\underline{EmpID}$, BillingRate, Specialty, FirmName)
- **DEPARTMENT** ($\underline{DeptID}$, DeptName, Phone, $ManagerID_{FK}$)
- **PROJECT** ($\underline{ProjID}$, ProjName, Location, Budget, $DeptID_{FK}$)
- **WORKS_ON** ($\underline{EmpID}_{FK}, \underline{ProjID}_{FK}$, HoursWorked, Role)
- **DEPENDENT** ($\underline{EmpID}_{FK}, \underline{DepName}$, Relationship, DOB)